

involves significant manual effort. Further, tests are often extremely long and targeted, running for several hours on silicon.

Random and constrained-random tests. In addition to focused tests, exercise systems feature through random and constrained-random testing. Examples of such tests include executing a random sequence of system instructions and exercising concurrent interleaving. The goal is to exercise the system in ways not conceived by humans. Random instruction tests can include hundreds of millions of random seeds generating instruction sequences.

In addition to the tests, their applications require development of specialised peripherals, boards and test cards. This is specifically pertinent for compatibility validation where the system needs to be exercised for a large number of peripheral devices, software versions and platform features.

Towards standardisation of validation infrastructure

Two such standardisations are ARM Coresight™ and Intel® Platform Analysis Tool. Both these architectures include a set of hardware IPs (for example, for collecting, synchronising and time-stamping signal traces and other observability collaterals from different design blocks, routing these to output ports and system memory, and software APIs for configuration, triggering, transport and analysis). Specifics of the architectures vary.

Coresight architecture is instantiated into macro-cells that can interact with IP functionality through a standard interface. Platform Analysis Tool includes a specialised IP called Trace Hub, which is responsible for the aggregation and transport of both hardware and software traces, together with APIs that enable direct interaction with this IP for transport and analysis.

While such standardisation assists in streamlining post-silicon readiness development, it must be emphasised that the current state of the art in standardisation is rather rudimentary. For instance, software tools for extracting trace data for both architectures

above are typically APIs for accessing different internal design collaterals.

Little assistance is provided to identify the specific collateral, which would be useful or profitable for debug purposes. It is left to the expertise of the human designer and validator to hook up the APIs with the hardware and software content in the target design for achieving validation objectives.

Trace signal selection

Trace signals are used to address the observability limitation during post-silicon debug. The idea is to trace a set of signals during run time and store in a trace buffer. This is done so that traced values can be used during post-silicon debug. Since I/O speed (for example, using JTAG) is significantly slower than the speed of execution (for instance, MHz versus GHz), it is not possible to dump the traced values through I/O ports during execution. Therefore an internal trace buffer is required.

Trace signal selection needs to maintain various design constraints. For example, trace buffer size directly translates to area and power overhead. Moreover, routing selected signals to the trace buffer may cause congestion and other layout-related issues. As a result, in a design with millions of signals, a typical trace buffer traces a few hundred signals for a few thousand cycles. For example, a 128 _ 2048 trace buffer can store 128 signals over 2048 clock cycles.

Design overhead considerations directly impose two constraints:

1. How to select a small number of trace signals that can maximise observability
2. How to effectively utilise traced values for a small number of cycles to enable meaningful debug

In post-silicon debug, unknown signal states can be reconstructed from the traced states in two ways: forwards and backwards restoration.

Pre-silicon stimuli generation

A pre-silicon stimuli generator has to provide the user with the ability to specify the desired scenarios in some convenient way and produce many valid high-quality test-cases according to the user's specification(s). **EFY**

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**In good times
 people want
 to advertise.
 In bad times
 they have to.**
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-Bruce Barton

